Day 1

From Wound Swab to Genome

Genomic Surveillance of Surgical Site Infections & AMR
08:30–17:00

Course files: github.com/aryakaul/ughe-ssi-genomics-course

Day 1 · Morning

Time	Session
08:30	Welcome and knowledge check
09:00	**Lecture 1:** SSIs, AMR and why genomics?
09:45	**Lecture 2:** From sample to sequence
10:30	*Break*
10:45	**Lab 1:** Files, formats and your first Galaxy history
12:15	*Lunch*

Day 1 · Afternoon

Time	Session
13:00	**Lecture 3:** How bacteria resist antibiotics
13:45	**Lab 2:** Quality control and genome assembly
15:30	*Break*
15:45	**Lab 3:** A look under the hood: the command line
16:45	Recap and exit ticket

Your course files

github.com/aryakaul/ughe-ssi-genomics-course

- **Lab handouts:** the `labs/pdf/` folder
- **Slides:** the `slides/pdf/` folder
- **Everything at once:** green **Code** button → **Download ZIP**

Before we start

- Log in to **usegalaxy.eu** and **pathogen.watch**
- Can’t log in? **Raise your hand now**: account emails can take a few minutes

WORKING IN PAIRS

Driver: hands on the keyboard · **Navigator:** reads the handout
You swap at each lab. Questions any time: **raise your hand**.

Choose a code word

- Pick a **code word** you'll remember, e.g. a favourite food or place
- You'll use the **same code word** tomorrow, so we can compare before and after
- No names: the knowledge check is anonymous

Knowledge check · Pre

- **Not graded**, and anonymous
- On your quiz sheet or a piece of paper, write your **code word** and **Pre**
- For each question, write its number and your answer (e.g. “1 b”)
- One question at a time on the screen

KEY MESSAGE

It’s fine not to know an answer. Just leave it blank.

KNOWLEDGE CHECK · QUESTION 1 OF 15

Which question can whole-genome sequencing answer that standard culture + AST cannot?

a) Which antibiotic should this patient receive today

b) Whether isolates from several patients are the same strain

c) Whether the organism is Gram-positive

d) Whether the patient has a fever

KNOWLEDGE CHECK · QUESTION 2 OF 15

Do you need to own a sequencer to do genomic surveillance? (Yes / No) Briefly explain why.

Write your answer on your paper

KNOWLEDGE CHECK · QUESTION 3 OF 15

Name two bacterial species that commonly cause surgical site infections.

Write your answer on your paper

KNOWLEDGE CHECK · QUESTION 4 OF 15

An Illumina sequencing run typically produces reads that are:

a) 100–300 bases long and highly accurate

b) 10,000+ bases long

c) the whole genome in one read

d) protein sequences

KNOWLEDGE CHECK · QUESTION 5 OF 15

Which technology is better at showing whether a resistance gene is on a plasmid?

a) Illumina short reads alone

b) Oxford Nanopore long reads

c) AST

d) MLST

KNOWLEDGE CHECK · QUESTION 6 OF 15

A genome is 5 million bases. You have 1 million reads of 150 bases. What is the approximate coverage?

a) 3×

b) 30×

c) 150×

d) 300×

What does a FASTQ file contain that a FASTA file does not?

a) Sequence names

b) DNA letters

c) Quality scores

d) Gene names

KNOWLEDGE CHECK · QUESTION 8 OF 15

After assembly, which result usually means a *better* assembly?

a) More contigs and a smaller N50

b) Fewer contigs and a larger N50

c) A total length twice the expected genome size

d) No contigs

The gene *mecA* in *S. aureus* indicates:

a) Vancomycin resistance

b) MRSA (methicillin resistance)

c) Carbapenem resistance

d) A virulence toxin

KNOWLEDGE CHECK · QUESTION 10 OF 15

blaNDM-1 is an example of:

a) An ESBL

b) A carbapenemase

c) An efflux pump

d) A housekeeping gene

KNOWLEDGE CHECK · QUESTION 11 OF 15

The genome shows no known resistance gene, but AST says “resistant”. Give one possible explanation.

Write your answer on your paper

Two *Klebsiella* isolates are ST15 and ST307. Could they be part of the same recent transmission chain?

a) Yes, very likely

b) No, they are different strain types

c) Only if they are both ESBL

KNOWLEDGE CHECK · QUESTION 13 OF 15

On a phylogenetic tree, what tells you how genetically different two isolates are?

a) Their order from top to bottom

b) The branch lengths between them

c) The colour of the labels

KNOWLEDGE CHECK · QUESTION 14 OF 15

Four isolates are 2–6 SNPs apart and the patients overlapped on the same ward. This most strongly suggests:

a) Unrelated infections

b) Possible transmission / outbreak

c) Contamination of the sequencer

KNOWLEDGE CHECK · QUESTION 15 OF 15

List three pieces of metadata that must be recorded with every isolate for surveillance.

Write your answer on your paper

How confident are you?

Rate each from **1** (not at all) to **5** (very confident):

- I can explain what a genome sequence can tell us about an SSI
- I can run a tool in Galaxy
- I can identify resistance genes from a genome
- I can interpret a phylogenetic tree
- I can plan a genomic surveillance project

N1 · About you

Your main role:
Clinician · Nurse / midwife · Laboratory · Public health / IPC · Research · Other

Your institution:
UGHE · Butaro Hospital · Other

N2 · At your site, is there…

Answer **Yes**, **No** or **Not sure**:

1. culture and antibiotic susceptibility testing?
2. long-term storage of bacterial isolates (e.g. a freezer)?
3. internet reliable enough to use web tools like Galaxy?

N3

What is the biggest barrier to using genomic data in your work?

Lecture 1

Surgical Site Infections, AMR and Why Genomics?

Day 1 · 09:00–09:45 · 45 min
≈ 30 min talk + 15 min discussion

LO1: explain how WGS strengthens SSI and AMR surveillance compared with culture and AST alone

Think of the last surgical patient whose infection didn’t respond to first-line antibiotics.

What did we learn about where it came from?

What is a surgical site infection?

- An infection at or near the incision
- Within **30 days** of surgery (**90 days** with an implant)
- Three levels:
  1. Superficial incisional
  2. Deep incisional
  3. Organ / space

The burden

- The **most common healthcare-associated infection** in low- and middle-income countries (WHO)
- Post-caesarean SSI is a major problem in sub-Saharan Africa, including rural Rwanda
- Consequences:
  - longer hospital stays and reoperation
  - catastrophic costs for households
  - death

SSI after caesarean section in Rwanda

10.9%
Kirehe District Hospital
Nkurunziza et al. 2019

5.7%
Kirehe, later cohort
Velin et al. 2021

6.85%
National pooled estimate
Sibomana et al. 2024

Resistance in SSI isolates from Kirehe

68%
of isolates were Gram-negative

0%
of Gram-negatives were susceptible to ampicillin

92%
were intermediate or resistant to **ceftriaxone**

Velin et al., *Ann Glob Health* 2021. Ceftriaxone is the standard empiric drug.

The gap

- Culture and AST capacity is still limited (Umutesi et al. 2021)
- We **don’t know which resistance genes or clones** are behind these numbers

KEY MESSAGE

That is the gap genomics fills.

DISCUSS

What fraction of SSIs in our hospitals get a culture? An AST?

AMR: the global picture (2019)

1.27 M
deaths **directly caused**
by bacterial AMR

4.95 M
deaths **associated**
with bacterial AMR

- The highest death rate was in **sub-Saharan Africa**
- 5 of the 6 leading pathogens are common causes of SSI

GRAM study, *Lancet* 2022

SSI pathogens to know

Pathogen	Resistance to watch
Staphylococcus aureus	MRSA (*mecA* / *mecC*)
Escherichia coli	ESBLs (CTX-M), carbapenemases
Klebsiella pneumoniae	ESBLs, carbapenemases (KPC, NDM, OXA-48)
Acinetobacter baumannii	Carbapenemases (OXA-23)
Pseudomonas aeruginosa	Efflux, porin loss, VIM / IMP

ESKAPE: *Enterococcus*, *S. aureus*, *Klebsiella*, *Acinetobacter*, *Pseudomonas*, *Enterobacter*

What our current tools tell us

CULTURE AND AST ANSWER

- **Which organism?**
- **Which drugs work?**

Essential for treating **this patient**.

THEY CAN'T ANSWER

- Are these 5 patients infected with the **same strain**?
- Is resistance spreading on a **plasmid** between species?
- Is a known **high-risk clone** circulating?

What a genome adds

A bacterial genome is 2–6 million letters (A, C, G, T). From one genome we get:

1 · Species
confirms or corrects the lab ID

2 · Strain type
MLST: “the bacterium’s surname”

3 · AMR genes
and **where** they sit: chromosome or plasmid

4 · Virulence genes
toxins and other factors

5 · Relatedness
SNP distance: the basis of outbreak detection

Case example: an outbreak response

MRSA on a **neonatal unit** in Cambridge, UK

- **Köser et al., *NEJM* 2012:** separated the linked outbreak isolates from unrelated MRSA that looked identical on routine lab tests
- **Harris et al., *Lancet Infect Dis* 2013:** traced transmission into the community and found a **staff carrier**, which ended the outbreak

KEY MESSAGE

Genomics changed the infection-control decision. We'll use this reasoning in Lab 6.

Genomic surveillance ≠ owning a sequencer



- A reference lab or service provider can do the **sequencing**
- The **expertise stays local**: choosing isolates, recording metadata, analysing, interpreting, communicating
- Africa CDC and national labs are building this network

The next two days

DAY 1 · THE DATA

What sequencing produces, and how to turn it into a genome

DAY 2 · THE ANSWERS

Resistance, strain types, outbreaks, and a sustainable programme

Discussion

15 min · pairs, then the whole room

1. Describe one situation in your work where knowing **“are these infections linked?”** would have changed what you did.
2. What happens to bacterial isolates in our labs after AST is done? **Are they stored?**

Lecture 2

From Sample to Sequence

Day 1 · 09:45–10:30 · 45 min

LO2: describe Illumina and Nanopore outputs and choose between them · LO6 (part): understand the outsourcing workflow

The journey of an isolate

Wound swab → **Culture** → **Pure isolate**

DNA extraction → **Library prep** → **Sequencing** → **Data files** → **Analysis**

KEY MESSAGE

Everything up to the pure isolate (dark boxes) already happens in our labs.

We sequence isolates, not swabs

- An **isolate** is a pure culture of one bacterium, so its genome is clean
- Sequencing a swab directly (**metagenomics**) mixes human DNA with many microbes: harder and more expensive
- For SSI surveillance, **isolate sequencing is the standard**

KEY MESSAGE

Store your isolates (–80 °C or glycerol stocks). You can’t sequence what you threw away.

DNA extraction and library preparation

- **Extraction:** break open the cells and purify the DNA
- **Library prep:** cut or tag the DNA and add short “adapter” sequences so the machine can read it
- **Multiplexing:** many samples are barcoded and run together, which lowers the cost per genome

> You don’t need to do this yourself. But poor-quality DNA gives poor-quality data (Lab 2).

Two main technologies

	Illumina (short reads)	Oxford Nanopore (long reads)
Read length	100–300 letters, in pairs	thousands to 100,000+ letters
Accuracy	very high	now high with modern chemistry
Best for	SNP-level outbreak work	complete genomes, **plasmids**
Machine	large, central labs	MinION is pocket-sized
Files	FASTQ pair (R1 + R2)	one FASTQ

Jigsaw analogy: Illumina gives many tiny pieces; Nanopore gives fewer, bigger pieces.

What is a read?

- A **read** is one piece of DNA sequence produced by the machine
- One bacterial genome gives **hundreds of thousands to millions** of reads
- We put the reads back together into the genome

> Like rebuilding a shredded book from many overlapping copies.

Coverage (depth)

Coverage = how many reads, on average, cover each position

$$(\text{number of reads} \times \text{read length}) \div \text{genome size}$$

$$1{,}000{,}000 \times 150 \div 5{,}000{,}000 = \mathbf{30\times}$$

KEY MESSAGE

Aim for ≥ 30× with Illumina and ≥ 40–50× with Nanopore.

Which should I use?

Question	Best choice
Is this an outbreak? (fine SNP differences)	Illumina
Is resistance on a plasmid moving between species?	Nanopore or hybrid
Need results fast, locally	Nanopore
Lowest cost per genome at large scale	Illumina (batched)
Best of both	**Hybrid** assembly

KEY MESSAGE

Your sequencing partner may use either. This course teaches both.

Outsourcing: how it works

Agree on a partner → **Ship isolates or DNA** → **Partner sequences** → **You analyse**

- **Partners:** national reference lab, Africa CDC network, academic collaborator, or a commercial service
- **Shipping:** follow their biosafety rules; you may need permits and a material transfer agreement
- **You get back:** FASTQ files, sometimes assemblies and reports. **Keep the data.**

Questions to ask a sequencing partner

Platform?
Illumina, Nanopore, or both

Coverage?
≥ 30× Illumina

Raw reads?
Make sure FASTQ files are included

Turnaround?
Weeks from shipping to data

Cost?
Per isolate, including shipping

Data sharing?
Who owns and publishes the data

Metadata: the most underrated part

A genome without **when, where and from whom** is almost useless for surveillance.

Minimum fields

- isolate ID · collection date
- facility · ward · specimen type
- surgery type and date
- AST results

PROTECT PRIVACY

Never put names or patient IDs in shared files. Use coded study IDs.

Standard: **PHA4GE** contextual data specification

Which sample, from where?

Field	Example for one SSI isolate
specimen collector sample ID	`UGHE-SSI-0001` (your lab's own ID)
specimen source ID	`CASE-0042`: coded, never a name
geo_loc name (country)	Rwanda
geo_loc name (state/province/region)	Northern Province

Who handled it?

Field	Example for one SSI isolate
sample collected by	Full official name of the hospital lab
sequenced by	Full official name of the sequencing partner
sequence submitted by	Who uploaded it to ENA / NCBI
data custodian organization name	Who decides how the data are used

Use full official names, not abbreviations.

When, what and why?

Field	Example for one SSI isolate
sample collection date	`2026-02-09` (YYYY-MM-DD, to the day)
organism	*Staphylococcus aureus*
purpose of sampling	Diagnostic testing
purpose of sequencing	Cluster/Outbreak investigation

Add your SSI fields

The 12 PHA4GE fields are the **minimum** for any pathogen. For SSI surveillance, add:

- ward · specimen type (e.g. wound swab)
- surgery type · surgery date · theatre
- AST results

KEY MESSAGE

Same column names in every hospital = data you can combine.

What files come back?

File	What it is
`sample_R1.fastq.gz` + `sample_R2.fastq.gz`	Illumina reads (a pair)
sample.fastq.gz	Nanopore reads
sample.fasta	Assembly (sometimes)
QC report	Sometimes

> In Lab 1 we'll open these files and look inside.

Quick check (show of hands)

1. You want to know if **6 MRSA isolates from one ward** are an outbreak. Which data would you request?

2. You suspect a **carbapenemase gene is jumping** between *Klebsiella* and *E. coli*. Which data would you request?

Quick check: answers

1. **Illumina, ≥ 30×**, with collection dates and ward metadata

2. **Nanopore or hybrid**, to resolve the plasmids

Break

Back at 10:45 for Lab 1

GENOMIC SURVEILLANCE OF SURGICAL SITE INFECTIONS & AMR · UGHE–HMS

Lab 1

Files, Formats and Your First Galaxy History

Day 1 · 10:45–12:15 · 90 min
Pairs: driver + navigator · swap at Part C

LO2, LO3: recognise genomic file formats, read a quality score, and run a tool in Galaxy

Open your handout

Lab 1 handout

`Lab1_files_formats_galaxy.pdf`

- **Download:** github.com/aryakaul/ughe-ssi-genomics-course
- Open the `labs/pdf/` folder and click the file

> The handout has every step. Copy reads, URLs and commands from the **PDF**, not from this screen.

Plan for this lab

Part	What	Time
A	Files and folders	15 min
B	Looking inside the formats	25 min
C	Your first Galaxy history	40 min
	Wrap-up	10 min

Before you start

- Log in to `usegalaxy.eu`
- **Driver:** hands on the keyboard
- **Navigator:** reads the handout and spots mistakes

> Stuck or have a question? **Raise your hand** and the instructor will come to you.

What's in a file name?

`isolate01_R1.fastq.gz`

`isolate01`	`R1`	`.fastq`	`.gz`
sample name	read 1 of a pair	format	compressed

CHECK YOURSELF

What do you expect is inside `UGHE-SSI-0042_R2.fastq.gz`?

Two more things to know

- **Plain-text files** (FASTA, FASTQ, CSV) are just letters: any text editor can open them
- **Genomic files are big**: 200 MB–1 GB per isolate

KEY MESSAGE

That’s why we analyse in Galaxy, on computers on the internet, not on our laptops.

FASTA: just sequence

```
>contig_1 length=120
TTCGCTCTATTGACTACGACGCGCTCATTCCCTTGTCGGAGAGTTATGGAACAAGGACGC
TGTCTGAGACTAGAAGACAGATAGTGCACACGACCGGCGTCGGAGAAACTCTATTTGCCG
>contig_2 length=60
CCTGACAAGTCAATGCGATCCGTAGGGGCAGCGCAGTATGCCAAGACTATAGGCACTGTC
```

- Each entry starts with > and a **name**; the sequence follows
- **Assemblies** (genomes) are stored in FASTA

CHECK YOURSELF

How many sequences? How many bases in total?

FASTQ: sequence + quality (raw reads)

Every read takes **exactly 4 lines**:

Line	Example	Meaning
1	`@SSI_isolate_01_read1`	@ + read name
2	`GCTAAAGACAATTACATAACATACACGTCAGCACGAAACT`	the DNA sequence
3	`+`	separator
4	`GHHGFGHIIIIIFIGGHGIFIIGHIHIGHIIGFHHDACBC`	quality, one symbol per base

Reading a quality score

Symbol	#	+	5	?	I
Q score	2	10	20	30	40
Chance it's wrong	≈ 63%	1 in 10	1 in 100	1 in 1,000	1 in 10,000

RULE OF THUMB

Capital letters (A–J) and ? or @ are good (Q30 and above).
Symbols like #, $, %, & are bad (below Q10).

Check yourself: read 4

```
IIHFGIIIFIIIIIFIIHFIIIIHI#'&'##&%''''$(%
```

CHECK YOURSELF

1. Which part of read 4 (start or end) has poor quality?
2. What should we do with those bases?

CSV / TSV: tables

```
isolate_id,organism,collection_date,ward
UGHE-SSI-0001,Klebsiella pneumoniae,2026-03-02,Maternity
UGHE-SSI-0002,Escherichia coli,2026-03-05,Surgical
```

- One row per isolate, one column per piece of information
- Commas (CSV) or tabs (TSV) separate the columns
- They open in Excel or Google Sheets

Swap driver and navigator

The navigator takes the keyboard; the driver reads the handout

A tour of Galaxy

- **Left, Tools:** search box and tool list
- **Centre:** tool forms and results
- **Right, History:** your files (**datasets**), in order

Colour	Meaning
green	finished
yellow	running
grey	queued
red	error

Create a history and upload data

1. History panel → **“+”** → rename it `Lab1 – formats`
2. **Upload** → **Paste/Fetch data** → paste the **4 reads** from your handout
3. **Type:** `fastqsanger` · **Name:** `tiny_reads.fastq`
4. **Start** → **Close**
5. When it turns **green**: click the name, then the **eye icon**

PART C4

Run your first tool: count lines

1. Search the tools: **Line/Word/Character count**
2. “Text file” → `tiny_reads.fastq`
3. **Run tool**, then view the result (eye icon)

> **CHECK YOURSELF**
>
> How many lines? How many reads is that? *(4 lines per read)*

PART C5

Run a real QC tool: FastQC

1. Search the tools: **FastQC** → input `tiny_reads.fastq` → **Run tool**
2. Open the **Webpage** output (eye icon)
3. Find **“Per base sequence quality”**

CHECK YOURSELF

- Does quality go up or down towards the end of the reads?
- Would you trust a report based on only 4 reads?

PART C6

Your history is a lab notebook

- Click a dataset, then the **“i”** (Dataset details) icon
- Galaxy records the **exact** tool, version and settings

KEY MESSAGE

This is reproducibility: anyone can see and repeat what you did.

Wrap-up

Format	Holds
FASTA	sequences (assemblies)
FASTQ	reads + quality (raw data)
CSV / TSV	tables (metadata, results)

Upload → **Choose tool** → **Run** → **View**

After lunch: **real** reads from a bacterial isolate.

Adapted from the Galaxy Training Network “Introduction to Galaxy” and “Quality Control” tutorials (CC-BY 4.0).

Lunch

Back at 13:00 for Lecture 3

Lecture 3

How Bacteria Resist Antibiotics

Day 1 · 13:00–13:45 · 45 min

LO4 (part): understand the genetic basis of AMR so that genome results can be interpreted

Four ways to survive an antibiotic

1 · Destroy the drug
Enzymes such as β-lactamases cut penicillins and cephalosporins

2 · Change the target
The drug no longer binds (PBP2a in MRSA; *gyrA* mutations)

3 · Pump it out
Efflux pumps

4 · Keep it out
Fewer or changed porins in Gram-negatives

Where resistance comes from: two routes

ACQUIRED GENE

The bacterium picks up a **new gene** from another bacterium

mecA, *blaCTX-M-15*, *blaNDM-1*

Genome: is the gene present or absent?

CHROMOSOMAL MUTATION

A change in the bacterium's **own gene**

gyrA S83L (ciprofloxacin)

Genome: compare with a reference and look for specific mutations

Plasmids: why resistance spreads so fast

- **Plasmids:** small circles of DNA that move **between bacteria, even between species**
- **Transposons and integrons:** “cut-and-paste” elements that move genes around
- One plasmid can carry **many** resistance genes at once

KEY MESSAGE

A resistant plasmid can spread even when the strain is not spreading. Genomics can tell these apart.

The big four to recognise

mecA
MRSA: resistant to almost all β-lactams

***blaCTX-M* (ESBL)**
Resistant to 3rd-generation cephalosporins such as ceftriaxone

Carbapenemases
KPC, NDM, OXA-48, VIM, IMP, OXA-23: resistant to carbapenems, the “last line”

***gyrA* / *parC* mutations**
Fluoroquinolone (ciprofloxacin) resistance

You don’t need to memorise gene names. The tools report them, and the glossary has a cheat sheet.

Other mechanisms you'll see

Genes	Resistance to
blaSHV, *blaTEM* (ESBL variants)	3rd-generation cephalosporins
aac, *aph*, *ant*, *armA*, *rmtB*	Gentamicin, amikacin
sul1, *sul2*, *dfrA*	Co-trimoxazole
tet(A), *tet(M)*	Tetracyclines

Reading gene names

blaCTX-M-15

bla	CTX-M	15
β-lactamase	family	variant

- aac(6')-Ib-cr: aminoglycoside enzyme; “cr” = also affects ciprofloxacin
- mecA: methicillin resistance

1-MINUTE EXERCISE

What do you predict blaNDM-1 does?

Can the genome predict the AST result?

WORKS VERY WELL

- *S. aureus*
- β-lactams in *E. coli* and *Klebsiella*

WORKS LESS WELL

- Resistance depends on how much a gene is **expressed**
- **Combinations** of weak mechanisms
- **Unknown** mechanisms not yet in databases

KEY MESSAGE

The genome complements AST. It does not replace AST for treating patients.

When the genome and the AST disagree

Genome says	AST says	Possible explanations
Resistance gene	Susceptible	Gene not expressed or broken · lab error · mixed culture
No gene found	Resistant	Unknown mechanism · mutation not in database · low coverage · mixed culture

> Disagreements are **useful**: they point to new mechanisms or lab QC problems. We'll practise this in Lab 4.

AMR tools and databases (preview of Lab 4)

Tool	From	Notes
AMRFinderPlus	NCBI, USA	Genes + mutations; organism-aware
ResFinder / PointFinder	CGE, Denmark	Web tool; predicts phenotype
CARD / RGI	McMaster, Canada	Very comprehensive
ABRicate	Community	Fast screening, several databases

> Tools may disagree slightly. Always report which tool and database version you used.

Summary

- Resistance = **genes** (often mobile) + **mutations**
- Plasmids let resistance spread **between species**
- Genome predictions are powerful but have limits: **always read them next to AST**

> Next: let’s make some genomes (Lab 2).

Lab 2

Quality Control and Genome Assembly

Day 1 · 13:45–15:30 · 105 min
Pairs · half the room on the Illumina track, half on the Nanopore track

LO3: perform QC and assembly on raw reads in Galaxy, and judge whether an assembly is good enough to use

Open your handout

Lab 2 handout

`Lab2_qc_assembly.pdf`

- **Download:** github.com/aryakaul/ughe-ssi-genomics-course
- Open the `labs/pdf/` folder and click the file

> The handout has every step. Copy reads, URLs and commands from the **PDF**, not from this screen.

One isolate, sequenced twice

- An **MRSA** strain (KUN1163), sequenced on **Illumina** and on **Oxford Nanopore**
- Half the pairs take each track; at the end you compare

STRATEGY

Assembly takes **20–60 min**. Start it early and interpret QC while it runs.
Not finished by **15:00**? Use the fallback (Part E).

Plan for this lab

Part	What	Time
A	Set up and upload	10 min
B–D	Check quality → clean → start assembly	35 min
F	Interpret QC while it runs	20 min
G	Assess the assembly	15 min
H	Compare tracks	10 min

Upload your track's reads

1. New history: `Lab2 – Illumina` **or** `Lab2 – Nanopore`
2. **Upload** → **Paste/Fetch data** → paste the URL(s) **from your handout**
3. **Type:** `fastqsanger.bz2` → **Start**
4. Rename: `illumina_R1` + `illumina_R2`, **or** `nanopore_reads`

> **CHECK YOURSELF**
>
> Why are there two Illumina files but only one Nanopore file?

Illumina: check, clean, assemble

FastQC → **fastp** → **Shovill**

1. **FastQC** on `illumina_R1`, then on `illumina_R2`
2. **fastp:** Paired · R1 and R2 · defaults
3. **Shovill:** Paired-end · the two fastp outputs · defaults

> Shovill can’t see the fastp files? Pencil icon → **Datatypes** → `fastqsanger.gz`

Nanopore: check, filter, assemble

NanoPlot → **filtlong** → **Flye**

1. **NanoPlot** on `nanopore_reads`
2. **filtlong:** Min. length `1000` · defaults
3. **Flye:** filtlong output · Mode **Nanopore corrected** · defaults

Assembly running?

Don’t wait for it: go on to Part F and interpret your QC

20 min

Interpret your QC

	Your track
Number of reads	
Typical read length	
Total bases (≈ reads × length)	
Coverage = total bases ÷ 2.8 Mb	
Quality good enough? Why?	

Check yourself: quality

CHECK YOURSELF

1. Is your coverage above the minimum?
   (≈ 30× Illumina, ≈ 40–50× Nanopore)
2. Look at the report from a **bad** run on the screen. What looks different?

Assess the assembly with QUAST

Quast → your Shovill `contigs` or Flye `assembly` → defaults → HTML report

Metric	Good sign for *S. aureus*
# contigs (pieces)	Illumina: tens to ≈ 100 · Nanopore: 1–5
Total length	≈ 2.7–3.0 Mb
N50	Bigger is better
GC (%)	≈ 33%

Check yourself: assembly

CHECK YOURSELF

1. Is the total length close to the expected genome size?
2. What would a total length of **5.6 Mb** suggest?
3. Would you accept this assembly for AMR analysis? Why?

Compare the two tracks

	Illumina	Nanopore
# contigs		
Largest contig		
Total length		
N50		

- Which is more **complete**?
- Which is more **accurate**?
- Gene on a **plasmid**? Which would you prefer?
- What would a **hybrid** give you?

Still running at 15:00?

1. Open the shared history link (handout Part E):
   usegalaxy.eu/u/aryakaul/h/ughe-ssi-course-precomputed
2. Click **Import**
3. Use the Shovill or Flye assembly and its QUAST report
4. Continue from Part G

> Your own job **keeps running**. Check it tonight!

Summary

Illumina → many accurate contigs

FastQC → **fastp** → **Shovill** → **QUAST**

Nanopore → few long contigs (≈ complete genome)

NanoPlot → **filtlong** → **Flye** → **QUAST**

Tomorrow: **what is in this genome?**

Adapted from the Galaxy Training Network “Bacterial Genome Assembly” tutorials (Shovill, Flye) (CC-BY 4.0).

Break

Back at 15:45. First thing: start JupyterLab (Lab 3, Part A)

Lab 3

A Look Under the Hood: the Command Line

Day 1 · 15:45–16:45 · 60 min
Pairs · swap driver and navigator at Part C

LO3: understand what Galaxy does behind the buttons, and take a first step towards independent analysis

Open your handout

Lab 3 handout

`Lab3_command_line_taster.pdf`

- **Download:** github.com/aryakaul/ughe-ssi-genomics-course
- Open the `labs/pdf/` folder and click the file

> The handout has every step. Copy reads, URLs and commands from the **PDF**, not from this screen.

Why bother?

- Every Galaxy tool you clicked today is a **command-line program**
- A dozen commands let you look inside huge files and follow online tutorials

KEY MESSAGE

You don’t need to become a programmer. This is the most durable skill in the course.

Golden rules

- Commands are **case-sensitive**, and spaces matter
- **Enter** runs a command · ↑ repeats the last one
- **Tab** completes file names
- **Ctrl + C** stops a running command

Open a terminal inside Galaxy

1. Run the tool **Interactive JupyterLab Notebook** (no changes)
2. After 1–3 min: **User** → **Active InteractiveTools** → click the link
3. In JupyterLab: **File** → **New** → **Terminal**
4. You'll see a prompt ending in `$`. Type here.

> Not working? Use your laptop's **Terminal** (macOS) or **Git Bash** (Windows).

Where am I? Moving around

Command	What it does
`pwd`	**p**rint **w**orking **d**irectory: where am I?
`ls`	list the files here
`mkdir ssi_course`	make a new folder
`cd ssi_course`	go into a folder
`cd ..`	go up one level

PART B

Try it

Type each line, then press Enter:

```
pwd
ls
mkdir ssi_course
cd ssi_course
cd ..
pwd
cd ssi_course
```

CHECK YOURSELF

Draw a little “tree” of the folders you moved through.

Swap driver and navigator

The navigator takes the keyboard; the driver reads the handout

Download the reads

The same Illumina read 1 file you used in Lab 2:

```
wget https://zenodo.org/…/DRR187559_1.fastqsanger.bz2
ls -lh
```

- Copy the **full URL** from your handout (Part C1)
- `ls -lh` lists files with their sizes (K, M, G)

PART C2

Peek inside a compressed file

```
bzcat DRR187559_1.fastqsanger.bz2 | head -8
```

- `bzcat` unzips the file as it reads it
- `head -8` shows the first 8 lines
- `|` is a **pipe**: it sends one command's output into the next

CHECK YOURSELF

How many reads did you see? Point to the first read's quality line.

PART C3

Count the reads

FASTQ has 4 lines per read: **count lines, then divide by 4**

```
bzcat DRR187559_1.fastqsanger.bz2 | wc -l
```

CHECK YOURSELF

How many reads? Does it match “Total sequences” in your Lab 2 FastQC report?

PART C3

A trap

Someone suggests counting reads like this:

```
bzcat DRR187559_1.fastqsanger.bz2 | grep -c "^@"
```

It counts lines that start with `@`.

> **CHECK YOURSELF**
>
> Why could this give the **wrong** answer?
> *Hint: which quality score is @?*

PART C4

Search inside the file

How many reads contain a run of 10 A's?

```
bzcat DRR187559_1.fastqsanger.bz2 | grep -c "AAAAAAAAAA"
```

`grep` finds lines containing a pattern; `-c` counts them

From reads to genomes

Download the assembly (URL in your handout, Part D), then:

```
head -3 DRR187559_contigs.fasta
grep -c ">" DRR187559_contigs.fasta
```

CHECK YOURSELF

What does `grep -c ">"` count? Compare it with **# contigs** in QUAST.

Galaxy ↔ command line

In Galaxy…	…on the command line
Upload → Paste URL	`wget URL`
Eye icon	`head`, `less`
Line/Word/Character count	`wc -l`
History	`history` (try it!)

> When you finish: **User** → **Active InteractiveTools** → **Stop** JupyterLab.

Cheat sheet 1: moving around

`pwd`	where am I?
`ls` · `ls -lh`	list files · with sizes
`cd folder` · `cd ..`	go into a folder · go up
`mkdir name`	make a folder
`wget URL`	download a file
`history`	commands you have run

Cheat sheet 2: looking inside files

`head -n 8 file`	first 8 lines
`wc -l file`	count lines
`grep "text" file`	find lines containing text (`-c` counts them)
`bzcat · zcat`	read a .bz2 · .gz file without unzipping
`cmd1 | cmd2`	pipe: send cmd1's output into cmd2

Want more? Software Carpentry **Unix Shell** lesson: `swcarpentry.github.io/shell-novice`

Adapted from The Carpentries "The Unix Shell" and the GTN "CLI basics" tutorial (CC-BY 4.0).

Today we…

1. Saw why genomics matters for SSIs and AMR
2. Learned what sequencing produces: reads, FASTQ, coverage
3. Ran our first tools in Galaxy
4. Turned raw reads into a genome and checked its quality
5. Looked under the hood with the command line

Exit ticket

What is still unclear?

One question on a slip of paper. No names.

> I’ll answer them first thing tomorrow.

See you tomorrow at 08:30

Day 2: from genomes to surveillance action